Twin identity and $3q\Phi_3$-perturbation rule for σ_1-G1 cores at $n = 6$

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Abstract

Let $A_\lambda(q) \in \mathbb{Z}_{\geq 0}[q]$ denote the palindromic core of the staircase Bott–Samelson trace $\sigma_1(V_\lambda) = \mathrm{tr}(\Pi_q|_{V\lambda})$ on the Specht module V_λ, after extracting $q^{a_\lambda}(1+q)^{b_\lambda}$. We prove three polynomial identities about the cores at $n = 5$ and $n = 6$, organised around the key shape $\lambda = (3,1,1)$ whose core $A_{(3,1,1)}(q) = q^2 + 4q + 1$ acts as a multiplicative scaffold:

$$\textsc{Lemma (n=5, atom relations)}: \quad A_{(3,2)} = A^2_{(3,1,1)} + q\Phi_3, \qquad A_{(4,1)} = A^2_{(3,1,1)} - q\Phi_3,$$

$$\textsc{Theorem (n=6, twin pair)}: \quad A_{(3,2,1)} = A^3_{(3,1,1)} + 3q\Phi_3 \cdot A_{(3,1,1)},$$
$$A_{(5,1)} = A^3_{(3,1,1)} - 3q\Phi_3 \cdot A_{(3,1,1)},$$

where $\Phi_3(q) = q^2 + q + 1$ is the third cyclotomic polynomial. The three target identities of the proof session (Theorems A, B, C of `PROVE.md`) follow as immediate corollaries. In particular,

$$A_{(3,2,1)} + A_{(5,1)} = 2A^3_{(3,1,1)}, \qquad A_{(3,2,1)} - A_{(5,1)} = 6q\Phi_3 \cdot A_{(3,1,1)}.$$

The proof of each identity is a direct polynomial verification using the stair-basis σ_1-G1 expansion of [1]. The structural *origin* of the symmetric $\pm q\Phi_3$ pattern at $n = 5$, and of its exact triplication at $n = 6$, remains mysterious; we close with the precise combinatorial questions left open.

1 Setup

1.1 Hecke algebra and σ_1

Let H_q be the Hecke algebra of $\mathfrak{S}_n$ over $\mathbb{Z}[q, q^{-1}]$, with generators $T_1, \ldots, T_{n-1}$ subject to $T_i^2 = (q-1)T_i + q$. Fix the staircase reduced word for the longest element $w_0 \in \mathfrak{S}_n$:

$$\underline{w}_0 = (i_1, \ldots, i_N), \qquad N = \binom{n}{2},$$

and define the staircase Bott–Samelson product

$$\Pi_q := \prod_{j=1}^{N} (T_{i_j} + 1) \in H_q.$$

For each partition $\lambda \vdash n$ let V_λ be the (cell/Specht) module and

$$\sigma_1(V_\lambda) := \mathrm{tr}(\Pi_q|_{V\lambda}) \in \mathbb{Z}[q].$$

By the bar involution σ_1 is palindromic of degree $\leq N$ in q, and by Kazhdan–Lusztig positivity $\sigma_1 \in \mathbb{Z}_{\geq 0}[q]$. Let a_λ, b_λ be the maximal q- and $(1+q)$-divisibilities of $\sigma_1(V_\lambda)$, and define the σ_1-*G1 core*

$$A_\lambda(q) := \frac{\sigma_1(V_\lambda)}{q^{a_\lambda}(1+q)^{b_\lambda}} \in \mathbb{Z}[q].$$

1.2 The σ_1-G1 stair basis

Theorem 1 ([1], σ_1-G1)**.** *For every $\lambda \vdash n$ there exist non-negative integers $n_k(\lambda) \in \mathbb{Z}_{\geq 0}$ for $k \in \{0, 1, \ldots, N\}$ with $k \equiv N \pmod 2$, such that*

$$\sigma_1(V_\lambda) = \sum_{\substack{0 \leq k \leq N \\ k \equiv N \pmod 2}} n_k(\lambda)\,(1+q)^k\, q^{(N-k)/2}. \tag{1}$$

The numbers $n_k(\lambda)$ are obtained as a sum over closed length-N paths in the Kazhdan–Lusztig W-graph of V_λ, with k "stay" (descent) steps and $N-k$ "move" (KL-edge) steps.

Dividing (1) by $q^{a_\lambda}(1+q)^{b_\lambda}$ gives the *stair-basis expansion* of A_λ:

$$A_\lambda(q) = \sum_{i=0}^{r_\lambda} c_i(\lambda)\, q^{r_\lambda - i}\,(1+q)^{2i}, \qquad c_i(\lambda) := n_{b_\lambda + 2i}(\lambda), \tag{2}$$

where r_λ is the integer such that the support of $n_k(\lambda)$ is $\{b_\lambda, b_\lambda+2, \ldots, b_\lambda+2r_\lambda\}$. Equivalently $r_\lambda = (N - 2a_\lambda - b_\lambda)/2$. All $c_i(\lambda) \in \mathbb{Z}_{\geq 0}$ by Theorem 1. Two examples used throughout:

Example 2. At $n = 5$ ($N = 10$): each rank-1 atom has $r_\lambda = 2$, so the stair basis at $n = 5$ is $(q^2,\, q(1+q)^2,\, (1+q)^4)$. We have

$$A_{(3,2)} = 3\,q^2 + 5\,q(1+q)^2 + (1+q)^4, \qquad A_{(4,1)} = 5\,q^2 + 3\,q(1+q)^2 + (1+q)^4,$$

$$A_{(3,1,1)} = 2\,q + (1+q)^2 \quad (\text{at } r_\lambda = 1, \text{ basis } (q, (1+q)^2)).$$

Example 3. At $n = 6$ ($N = 15$): the rank-2 atoms $A_{(3,2,1)}$ and $A_{(5,1)}$ both have $r_\lambda = 3$, so the stair basis at $n = 6$ for them is $(q^3,\, q^2(1+q)^2,\, q(1+q)^4,\, (1+q)^6)$. We have

$$A_{(3,2,1)} = 2\,q^3 + 15\,q^2(1+q)^2 + 9\,q(1+q)^4 + (1+q)^6,$$

$$A_{(5,1)} = 14\,q^3 + 9\,q^2(1+q)^2 + 3\,q(1+q)^4 + (1+q)^6.$$

1.3 Five base atoms

The polynomials at the centre of this paper are:

$$\begin{aligned}
A_{(3,1,1)} &= q^2 + 4q + 1, \\
A_{(3,2)} &= q^4 + 9q^3 + 19q^2 + 9q + 1, \\
A_{(4,1)} &= q^4 + 7q^3 + 17q^2 + 7q + 1, \\
A_{(3,2,1)} &= q^6 + 15q^5 + 66q^4 + 106q^3 + 66q^2 + 15q + 1, \\
A_{(5,1)} &= q^6 + 9q^5 + 36q^4 + 70q^3 + 36q^2 + 9q + 1.
\end{aligned}$$

We write $\Phi_3(q) := q^2 + q + 1$ for the third cyclotomic polynomial.

2 The $\pm q\Phi_3$ pattern at $n = 5$

Lemma 4 ($n = 5$ atom relations)**.**

$$A_{(3,2)}(q) = A_{(3,1,1)}(q)^2 + q\,\Phi_3(q), \qquad A_{(4,1)}(q) = A_{(3,1,1)}(q)^2 - q\,\Phi_3(q).$$

Proof. Direct computation:

$$\begin{aligned} A^2_{(3,1,1)} &= (q^2 + 4q + 1)^2 = q^4 + 8q^3 + 18q^2 + 8q + 1, \\ q\,\Phi_3(q) &= q\,(q^2 + q + 1) = q^3 + q^2 + q. \end{aligned}$$

Adding: $A^2_{(3,1,1)} + q\,\Phi_3 = q^4 + 9q^3 + 19q^2 + 9q + 1 = A_{(3,2)}$. Subtracting: $A^2_{(3,1,1)} - q\,\Phi_3 = q^4 + 7q^3 + 17q^2 + 7q + 1 = A_{(4,1)}$. □

Corollary 5 ($n = 5$ symmetric averaging)**.**

$$A_{(3,2)} + A_{(4,1)} = 2\,A^2_{(3,1,1)}, \qquad A_{(3,2)} - A_{(4,1)} = 2\,q\,\Phi_3(q) =: \delta(q).$$

Remark 6 (Stair-basis interpretation of $q\Phi_3$). In the $n = 5$ stair basis $(q^2, q(1+q)^2, (1+q)^4)$, $A^2_{(3,1,1)}$ has coordinates $(4, 4, 1)$, while $A_{(3,2)}$ has $(3, 5, 1)$ and $A_{(4,1)}$ has $(5, 3, 1)$. The perturbation $q\Phi_3 = -q^2 + q(1+q)^2$ has stair coordinates $(-1, +1, 0)$, so $A_{(3,2)} - A^2_{(3,1,1)}$ flips -1 from q^2 into $+1$ for $q(1+q)^2$, and $A_{(4,1)} - A^2_{(3,1,1)}$ does the reverse. Lemma 4 is the statement that the rank-1 pair $\{(3, 2), (4, 1)\}$ at $n = 5$ is symmetric across $A^2_{(3,1,1)}$ along this $q\Phi_3$-axis.

3 The $\pm 3q\Phi_3$ pattern at $n = 6$ (twin pair)

Theorem 7 ($n = 6$ twin pair)**.**

$$A_{(3,2,1)}(q) = A_{(3,1,1)}(q)^3 + 3\,q\,\Phi_3(q) \cdot A_{(3,1,1)}(q),$$

$$A_{(5,1)}(q) = A_{(3,1,1)}(q)^3 - 3\,q\,\Phi_3(q) \cdot A_{(3,1,1)}(q).$$

Equivalently,

$$\frac{A_{(3,2,1)}(q)}{A_{(3,1,1)}(q)} = A_{(3,1,1)}(q)^2 + 3\,q\,\Phi_3(q), \qquad \frac{A_{(5,1)}(q)}{A_{(3,1,1)}(q)} = A_{(3,1,1)}(q)^2 - 3\,q\,\Phi_3(q).$$

Proof. Compute

$$A^3_{(3,1,1)} = (q^2 + 4q + 1)^3 = q^6 + 12q^5 + 51q^4 + 88q^3 + 51q^2 + 12q + 1.$$

And $3q\Phi_3 \cdot A_{(3,1,1)} = 3q(q^2 + q + 1)(q^2 + 4q + 1) = 3q(q^4 + 5q^3 + 7q^2 + 5q + 1) = 3q^5 + 15q^4 + 21q^3 + 15q^2 + 3q$. Adding,

$$A^3_{(3,1,1)} + 3q\Phi_3 A_{(3,1,1)} = q^6 + 15q^5 + 66q^4 + 106q^3 + 66q^2 + 15q + 1 = A_{(3,2,1)}.$$

Subtracting,

$$A^3_{(3,1,1)} - 3q\Phi_3 A_{(3,1,1)} = q^6 + 9q^5 + 36q^4 + 70q^3 + 36q^2 + 9q + 1 = A_{(5,1)}.$$ □

Corollary 8 (Twin sum and difference)**.**

$$A_{(3,2,1)} + A_{(5,1)} = 2\,A^3_{(3,1,1)}, \qquad A_{(3,2,1)} - A_{(5,1)} = 6\,q\,\Phi_3 \cdot A_{(3,1,1)} = 3\,\delta \cdot A_{(3,1,1)}.$$

Remark 9 (Stair-basis interpretation)*.* In the $n = 6$ stair basis $(q^3, q^2(1+q)^2, q(1+q)^4, (1+q)^6)$, $A^3_{(3,1,1)}$ has stair coordinates $(8, 12, 6, 1)$, while $A_{(3,2,1)}$ has $(2, 15, 9, 1)$ and $A_{(5,1)}$ has $(14, 9, 3, 1)$. The perturbation $3q\Phi_3 \cdot A_{(3,1,1)}$ has stair coordinates $(-6, +3, +3, 0)$, the Cauchy product of the $n = 5$ perturbation $3q\Phi_3 = (-3, +3, 0)$ and $A_{(3,1,1)} = (2, 1)$. The pair $\{(3, 2, 1), (5, 1)\}$ at $n = 6$ is symmetric across $A^3_{(3,1,1)}$ along an enlarged perturbation axis.

4 Theorems A, B, C of `PROVE.md` as corollaries

Corollary 10 (Theorem A, Twin identity)**.**

$$A_{(3,2,1)} + A_{(5,1)} = A_{(3,1,1)} \cdot \big(A_{(3,2)} + A_{(4,1)}\big).$$

Proof. By Corollary 8, $A_{(3,2,1)} + A_{(5,1)} = 2A^3_{(3,1,1)} = A_{(3,1,1)} \cdot 2A^2_{(3,1,1)}$. By Corollary 5, $2A^2_{(3,1,1)} = A_{(3,2)} + A_{(4,1)}$. □

Corollary 11 (Theorem B, Sign-flipped pair)**.** *With* $\delta(q) = A_{(3,2)} - A_{(4,1)} = 2q\Phi_3(q)$*,*

$$A_{(3,2,1)} = A_{(3,1,1)} \cdot (A_{(3,2)} + \delta) = A_{(3,1,1)} \cdot (2A_{(3,2)} - A_{(4,1)}),$$

$$A_{(5,1)} = A_{(3,1,1)} \cdot (A_{(4,1)} - \delta) = A_{(3,1,1)} \cdot (2A_{(4,1)} - A_{(3,2)}).$$

Proof. By Lemma 4, $A_{(3,2)} = A^2_{(3,1,1)} + q\Phi_3$, so $A_{(3,2)} + \delta = A^2_{(3,1,1)} + q\Phi_3 + 2q\Phi_3 = A^2_{(3,1,1)} + 3q\Phi_3$. Multiplying by $A_{(3,1,1)}$ and using Theorem 7 gives $A_{(3,1,1)} \cdot (A_{(3,2)} + \delta) = A^3_{(3,1,1)} + 3q\Phi_3 A_{(3,1,1)} = A_{(3,2,1)}$. The substitution $A_{(3,2)} + \delta = 2A_{(3,2)} - A_{(4,1)}$ follows from $\delta = A_{(3,2)} - A_{(4,1)}$ (Corollary 5). Symmetric argument for $A_{(5,1)}$. □

Corollary 12 (Theorem C, $2q$-perturbation rule)**.** $A_{(3,2,1)} = A_{(3,1,1)} \cdot \mathrm{atom}_{\mathrm{RTL}}$*, where* $\mathrm{atom}_{\mathrm{RTL}}(q) := q^4 + 11q^3 + 21q^2 + 11q + 1$ *satisfies*

$$\mathrm{atom}_{\mathrm{RTL}}(q) = A_{(3,2)}(q) + 2\,q\,\Phi_3(q) = A_{(3,1,1)}(q)^2 + 3\,q\,\Phi_3(q).$$

Proof. The left equality is Corollary 11 (substitute $A_{(3,2)} + \delta$). For the right equality, apply Lemma 4 again: $A_{(3,2)} + 2q\Phi_3 = A^2_{(3,1,1)} + q\Phi_3 + 2q\Phi_3 = A^2_{(3,1,1)} + 3q\Phi_3$. □

5 Computational verification

All polynomial identities above were verified by direct expansion in `sympy`; see `~/projects/scratch/2026-04-29-twin-identity-proof/verify.py`. Specifically:

- Lemma 4 (two identities): **OK**.
- Corollary 5 (sum, difference): **OK**.
- Theorem 7 (two identities): **OK**.
- Corollary 8 (sum $= 2A^3_{(3,1,1)}$, difference $= 6q\Phi_3 A_{(3,1,1)}$): **OK**.

- Corollary 10 (Theorem A): **OK**.
- Corollary 11 (Theorem B, both halves): **OK**.
- Corollary 12 (Theorem C, both equalities): **OK**.

The five base atoms have themselves been computed independently from the Hoefsmit/seminormal Hecke matrices (see `~/projects/scratch/2026-04-25-hu-mathas-test/compute_filtration.py` and `~/projects/scratch/2026-04-29-branching-decomp-other-shapes/compute.py`).

6 What is structural, what is mysterious

6.1 What is structural

Stair-basis expansion. The σ_1-G1 theorem ([1], recalled here as Theorem 1) gives every A_λ as a non-negative integer combination of stair monomials $q^{r-i}(1+q)^{2i}$, $0 \le i \le r$. This is a representation-theoretic statement: the integer $n_k(\lambda)$ counts closed length-N paths in the W-graph of V_λ with exactly k "stay" (descent) steps and $N-k$ "move" (KL-edge) steps; the coefficient of each path is a product of KL μ-coefficients and is non-negative by KL positivity.

Multiplicativity of stair basis. The stair monomials of degree r at n multiply with the stair monomials of degree r' at n' as $q^{r-i}(1+q)^{2i} \cdot q^{r'-j}(1+q)^{2j} = q^{(r+r')-(i+j)}(1+q)^{2(i+j)}$, i.e., the stair basis is closed under polynomial multiplication. This is the only structural fact used to lift identities from $n=5$ atoms to $n=6$ atoms: the polynomial product $A_{(3,1,1)} \cdot A_{(3,2)}$ is again a non-negative integer combination of stair monomials, in the stair basis at degree $r_{(3,1,1)} + r_{(3,2)}$.

Symmetric pair across a power of $A_{(3,1,1)}$. The empirical pattern is that the symmetric pair $\{(3,2),(4,1)\}$ at $n=5$ straddles $A^2_{(3,1,1)}$, and the symmetric pair $\{(3,2,1),(5,1)\}$ at $n=6$ straddles $A^3_{(3,1,1)}$. Each is a $\pm$ perturbation by an integer multiple of $q\Phi_3(q)$ times an appropriate power of $A_{(3,1,1)}$.

6.2 What remains mysterious

Question 13 (Origin of $q\Phi_3$). Why is the perturbation kernel $q\Phi_3(q) = q+q^2+q^3$? Both Φ_3 and the leading q are natural in the W-graph context (Φ_3 specialises the Hecke algebra at a primitive third root of unity, where new idempotents appear; q is one full "move" weight in the σ_1-G1 expansion). But we have not identified the specific closed W-graph paths whose generating function is $q\Phi_3$. In particular, the $n=5$ stair-basis decomposition $q\Phi_3 = -q^2 + q(1+q)^2 + 0 \cdot (1+q)^4$ involves a *negative* stair coefficient, which is necessarily a *signed* cancellation between paths in the W-graphs of $V_{(3,2)}$ and $V_{(4,1)}$, not a path-counting identity in either separately.

Question 14 (Origin of the factor 3). Why does the $n=5$ kernel $q\Phi_3$ *exactly triple* on passing from $n=5$ to the lifted $n=6$ pair, becoming $3q\Phi_3$ when measured against $A_{(3,1,1)}$ as multiplicative scaffold?

- The number of removable corners of $(3,2,1)$ at $n=6$ is 3 (cells $(1,3),(2,2),(3,1)$); this matches. But the same factor 3 also appears in $A_{(5,1)}/A_{(3,1,1)} = A^2_{(3,1,1)} - 3q\Phi_3$, and $(5,1)$ has only 2 removable corners. So "corners" is not the right invariant.

- Equivalently, in the stair basis, the kernel passes from $(-1,+1,0)$ at $n=5$ to $(-3,+3,0)$ at $n=5$-rescaled, and then is convolved with $A_{(3,1,1)}=(2,1)$ to produce the $n=6$ shift $(-6,+3,+3,0)$. What is the $n=6$ structural object that scales the kernel by 3?

Question 15 (Reason $A_{(3,1,1)}$ divides $A_{(5,1)}$). The divisibility $A_{(3,1,1)} \mid A_{(3,2,1)}$ is consistent with branching: $V_{(3,2,1)}{\downarrow}\mathfrak{S}_5 \supset V_{(3,1,1)}$. But $V_{(5,1)}{\downarrow}\mathfrak{S}_5 = V_{(5)} \oplus V_{(4,1)}$ does *not* contain $V_{(3,1,1)}$, yet $A_{(3,1,1)}$ still divides $A_{(5,1)}$. Hence $A_{(3,1,1)}$ plays a structural role at $n=6$ that transcends naive Hecke branching. We have no explanation for this beyond the polynomial identity.

Remark 16 (Specialisation at $q=\omega$, primitive third root of unity). Since $\Phi_3(\omega)=0$, both $q\Phi_3$ and $3q\Phi_3$ vanish at $q=\omega$. Therefore at $q=\omega$,

$$A_{(3,2)}(\omega) = A_{(4,1)}(\omega) = A_{(3,1,1)}(\omega)^2, \qquad A_{(3,2,1)}(\omega) = A_{(5,1)}(\omega) = A_{(3,1,1)}(\omega)^3.$$

A direct computation gives $A_{(3,1,1)}(\omega)=3\omega$, so $A_{(3,2,1)}(\omega)=A_{(5,1)}(\omega)=27\omega^3=27$. This collapse at $q=\omega$ is consistent with the perturbation pattern but, again, not explained by it.

6.3 What is the gap

A *structural* proof of Lemma 4 and Theorem 7 would identify, for each of the four directed perturbation polynomials

$$A_{(3,2)} - A_{(3,1,1)}^2 = +q\Phi_3, \quad A_{(4,1)} - A_{(3,1,1)}^2 = -q\Phi_3,$$

$$A_{(3,2,1)}/A_{(3,1,1)} - A_{(3,1,1)}^2 = +3q\Phi_3, \quad A_{(5,1)}/A_{(3,1,1)} - A_{(3,1,1)}^2 = -3q\Phi_3,$$

a precise object on which the difference is computed: a class of W-graph paths, a Soergel bimodule difference, an Ext-group, or a similar categorical or representation-theoretic invariant whose generating function is $\pm q\Phi_3$ at $n=5$ and $\pm 3q\Phi_3 A_{(3,1,1)}$ at $n=6$. We do not produce such an object here. The polynomial proofs above are honest identities, but they are not yet proofs at the W-graph or Hecke level.

7 Discussion

7.1 Relation to other rank-2 shapes at $n=6$

The third rank-2 shape at $n=6$ is $(4,1,1)$ (whose conjugate is $(3,1,1,1)$); we have not addressed it here because, computationally, $A_{(4,1,1)} = A_{(3,1,1)} \cdot A_{(3,1)\downarrow n=4} \cdots$ admits a different multiplicative structure. Likewise the rank-3 shape $(4,2)$ does *not* factor through $A_{(3,1,1)}$ multiplicatively (verified computationally: $A_{(3,1,1)} \nmid A_{(4,2)}$). The dichotomy "rank-2 factors through $A_{(3,1,1)}$, rank-3 does not" was identified earlier (Apr 26 work, $\lambda=(4,2)$ at $n=6$).

7.2 What this proof unlocks

Theorem 7, combined with the trace-square identity at $\lambda=(3,2,1)$ (Apr 29 work, `~/projects/scratch/2026-04-29-trace-square-321/`),

$$A_{(3,2,1)}^2 = G_{(3,2,1)} + 2q \cdot Q_{(3,2,1)} \quad \text{where } G_{(3,2,1)} = \sigma_2(V_\lambda)/q^8(1+q)^2,$$

and $Q_{(3,2,1)}$ is the deg-10 orphan atom $A_{10}(q)$, specialises to

$$A_{(3,1,1)}^2\,(\mathrm{atom}_{\mathrm{RTL}})^2 = G_{(3,2,1)} + 2q\,A_{10}.$$

Substituting $\mathrm{atom}_{\mathrm{RTL}} = A^2_{(3,1,1)} + 3q\Phi_3$ (Corollary 12):

$$A^2_{(3,1,1)}\left(A^2_{(3,1,1)} + 3q\Phi_3\right)^2 = G_{(3,2,1)} + 2qA_{10}.$$

This is a polynomial identity now expressed entirely in $n = 5$ atoms and the universal kernel $q\Phi_3$. It places A_{10} as the obstruction to σ_2-G1 inheritance at $\lambda = (3, 2, 1)$ admitting a closed-form expression in $n = 5$ atoms. We do not pursue this further here.

7.3 Stretch goal: generalisation to rank-2 pairs at all n

The $(3, 2)/(4, 1)$ pair at $n = 5$ and the $(3, 2, 1)/(5, 1)$ pair at $n = 6$ both satisfy

$$A_\lambda + A_{\lambda^*} = 2A^k_{(3,1,1)}$$

for $k = 2, 3$ respectively, where $A_{(3,1,1)} = q^2 + 4q + 1$. The conjecture

$$A_\lambda + A_{\lambda^*} = 2A^{k_n}_{(3,1,1)}$$

for k_n depending on n would generalise to an infinite family. Numerical evidence at $n = 7$ for the candidate pair $(4, 2, 1) \leftrightarrow (5, 1, 1)$ would falsify or confirm this. We have not tested this here.

References

[1] Clio, *Non-negativity of the trace core for the staircase Bott–Samelson operator on Hecke cell modules*, `~/projects/proofs/2026-04-24-sigma1-G1.tex`, 24 April 2026.

[2] D. Kazhdan and G. Lusztig, *Representations of Coxeter groups and Hecke algebras*, Invent. Math. 53 (1979), 165–184.

[3] B. Elias and G. Williamson, *The Hodge theory of Soergel bimodules*, Ann. of Math. (2) 180 (2014), 1089–1136.